Negotiated Agency in Generative User Interfaces

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Generative user interfaces (GenUI)—interfaces synthesized by AI models at design time or runtime—introduce a new condition for HCI: interaction structures may be produced through probabilistic inference rather than fully specified in advance. This shift reconfigures how user agency is structured and exercised. Drawing on foundational HCI accounts of agency, we identify core assumptions underlying traditional interaction paradigms: human-authored structures, legible system behavior, and stable interfaces. Runtime GenUI complicates these assumptions by delegating the formation of interaction structures to models. We propose *negotiated agency* as a lens for understanding this condition and outline implications for design and evaluation of model-centric systems.

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1 Agency as a Foundational Concept in HCI

User agency has long been central to HCI. Early paradigms grounded agency in direct manipulation: users act on visible objects, receive immediate feedback, and can reverse actions [13]. Norman similarly emphasized the importance of causal mental models linking intention, action, and outcome [12]. In these accounts, agency depends on legibility, predictability, and structural stability. Subsequent work expanded this understanding: Instrumental interaction framed agency as users' ability to compose and adapt interaction instruments [1]; end-user programming positioned agency as appropriation—the capacity to reshape computational artifacts [11]; more recently, malleable software critiques locked-down platforms and argues that users must be able to inspect and repurpose their tools to retain meaningful control [10]. Across these traditions, three assumptions recur: interaction structures are human-authored; system behavior is sufficiently legible to support stable mental models; and agency is exercised through interfaces that persist long enough to be learned and appropriated. These assumptions underpin prevailing usability heuristics and evaluation practices. Yet they also reflect a particular condition of computing: one in which interaction structures are designed in advance and remain relatively stable during use.

2 Generative UI as a Structural Shift

Generative user interfaces (GenUI) introduce a different condition. When AI models synthesize layouts, components, or workflows at runtime, interaction structures are not fully specified by human designers [9]. Designers configure prompts, constraints, or exemplars, but concrete interfaces emerge through probabilistic inference. Recent work illustrates this shift in both design-time tools [2, 8] and runtime systems that generate ephemeral or task-specific interfaces during

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interaction [3]. In such systems, interfaces are no longer fixed artifacts but are provisionally assembled in response to inferred intent.

This condition complicates the assumptions outlined above. Legibility becomes harder to establish because users must reason about not only interface state but also how the system interprets their input [7]. Controllability becomes indirect, mediated through prompts or corrections rather than explicit commands over system behavior. Stability weakens as interaction structures may reorganize across sessions or contexts. These shifts are not isolated breakdowns of existing properties, but reflect a relocation of where interaction is structured: from predefined interfaces to model-mediated generation. Users now engage with systems that continuously produce and adapt the interaction space itself. This reconfiguration raises a central question about how is user agency exercised under these conditions?

3 From User Control to Negotiated Agency

GenUI encompasses two scenarios. In design-time GenUI, models assist designers while users interact with stable artifacts, and agency remains largely consistent with traditional systems. A more substantial shift occurs in runtime GenUI, where interface structure is partially assembled during interaction. In these systems, users do not typically control the generative mechanism. They rarely access prompts or transformation rules, but influence generation indirectly through input, correction, and iterative interaction. Users steer outcomes without defining the generative process that shapes structural change.

This marks a shift from structural control to mixed-initiative interaction [5]. Agency becomes distributed across users, designers, and models rather than exercised unilaterally [14]. However, this redistribution is not equivalent to shared control. While such systems may feel empowering due to responsiveness, responsiveness should not be conflated with authority. Users may influence outputs without being able to contest underlying inferences, stabilize configurations, or override adaptation. Agency is thus experienced as negotiated: iterative yet remians bounded by model-defined constraints. Users participate in shaping outcomes, but only within a space of possibilities that they do not fully control. Understanding GenUI through negotiated agency foregrounds a central tension: how to support meaningful user participation in evolving interaction structures without reducing users to subjects of opaque inference.

4 Implications for HCI Design and Evaluation

If GenUI systems instantiate negotiated agency, then established approaches to design and evaluation must evolve accordingly. HCI should shift from analyzing stable interfaces to understanding how users engage with and influence dynamically generated interaction structures.

For HCI Design Practice. Design shifts from specifying interfaces to specifying conditions of generation. Designers must determine which aspects of interaction are open to inference, which remain fixed, and how users can intervene when outputs misalign with intent [9]. This also changes how design quality is understood. Heuristics such as consistency or predictability can no longer be evaluated at the level of single interface instances but must be considered across distributions of generated outcomes [15]. Participatory and value-sensitive approaches become essential when values are embedded in models rather than artifacts [4].

For Evaluation and Research Methods. GenUI also destabilizes traditional evaluation methods. Usability testing presumes comparable artifacts, whereas GenUI systems produce divergent interface configurations across users and moments. Evaluation must therefore shift from static artifacts to generative behavior. Longitudinal qualitative methods are well suited to studying how users appropriate evolving interfaces [6]. Quantitative metrics remain relevant insofar when capturing experiential constructs such as perceived agency, trust, and contestability rather than efficiency alone.

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